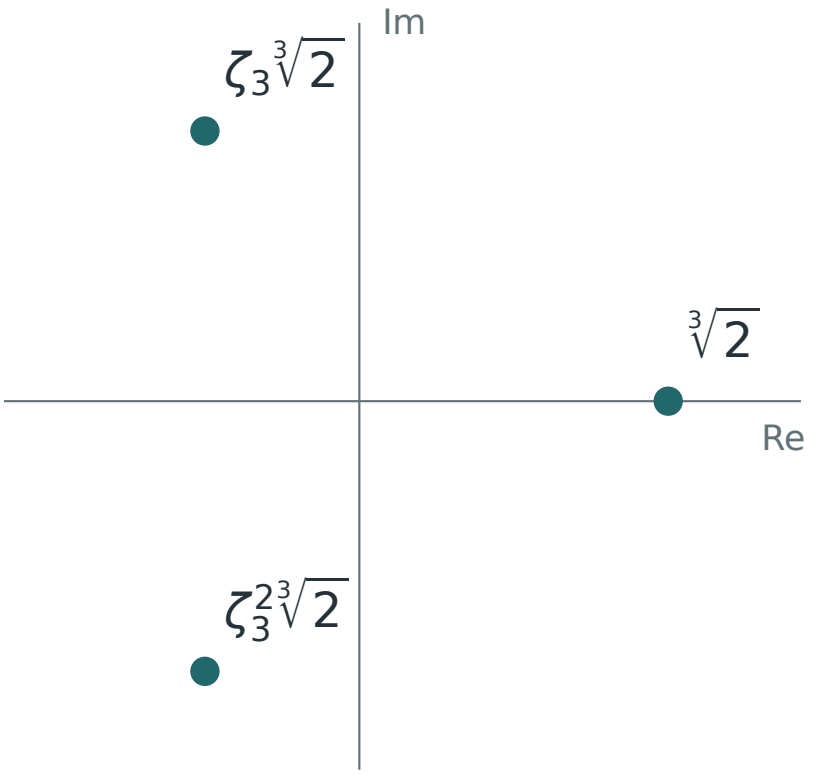
The splitting field needs the nonreal roots too

COMPUTATION / approximate positions, exact algebraic labels



$$L = \mathbb{Q}(\sqrt[3]{2}, \zeta_3)$$

$$[L : \mathbb{Q}] = 6$$

$$\Delta(x^3 - 2) = -108$$

$$\mathrm{Gal}(L/\mathbb{Q}) \cong S_3$$

The root plot illustrates the field. Irreducibility and discriminant certify the group.